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AI - CO1 - AT-3

① Soln :-

→ An intelligent agent preceives its environment through sensors & takes actions to achieve specific objectives.

① Simple Reflex agent :-

→ It Takes action based on the current input using condition-action rules. For Ex: It switches on lights when motion is detected & turns them off when no movement is detected. It is fast & simple.

② Model-Based agent :-

→ It maintains an internal model of the environment & uses previous information to make decisions. For Ex:- It can remember which rooms are occupied & adjust temperature accordingly. It performs better when environment changes.

③ Goal-Based agent :-

→ It selects actions that help achieve a particular goal. Ex:- It adjust the air conditioner to maintain a comfortable room-temperature. It is flexible but may require more computation to evaluate possible actions.

④ Utility - Based Agent :-

→ It chooses actions by comparing their benefits & selecting the option that provides the greatest overall satisfaction.

• Most effective Approach :-

→ A Hybrid approach is combining all these agents, supported by simple reflex rules, would be effective.

→ Thus, a hybrid intelligent agent provides better adaptability, energy management, user comfort & secure than using a single agent type.

② Soln :-

→ A robot navigating an unknown maze must explore paths until it reaches the Exit. Since, the layout is not known it can use uninformed search strategies

① Uniform Cost Search (UCS) :-

→ It expands the node with the lowest total cost from starting position. It uses a priority queue to select the next node. It finds the least path-cost.

• Advantage :- Finds an optimal solution when path cost are non-negative & the goal is reached.

• Disadvantages :- May Explore many nodes & consume considerable memory.

② Iterative Deepening Search (IDS):-

→ It repeatedly performs depth-limited search, increasing the depth limit each time until the exit is formed.

• Advantage:- Uses less memory & finds the shallowest solution when all steps have equal cost.

• Disadvantage:- Repeat the exploration of nodes at shallowes depths, causing some additional work.

• Most effective combination:-

→ A model based, goal-based agent using UCS is suitable when maze paths have different movement costs & finding the least-cost route is important.

→ If all movements have equal costs & memory is limited, IDS may be better choice.

→ ∴ The best combination depends on the maze movements cost, memory limitation & the robot's objectives.